MLB WIZARD PIPELINE

Shadow Ensemble v5.1 — Technical Documentation

April 20, 2026 · Updated with all recent methodology changes

9	20	97+	50K	3-Shot	RTX 5080
Models	Pipeline Steps	Features	MC Trials	Scheduler	GPU

What's new in this update (April 2026):

- **F5 (First 5 Innings) Model** — complete new XGBoost + Bayesian Hierarchical Stacker for the F5 +0.5 run line market
- **Enriched Feature Matrix v2** — 8 new feature groups: Lineup wRC+, SP FIP/LOB%, Barrel%, Run Diff, Team Metrics, DER, Elo Ratings, Catcher Framing
- **K-Prop Tracker** — strikeout prediction logging vs PrizePicks lines
- **Blend Tracker + Lock Optimizer** — rolling grid-search of optimal MC/XGB weights and gate constants
- **Expanded Scheduler** — now fires at 10:00 AM ET (moved from 11 AM), 20 pipeline steps (was 9)
- **Catcher Framing** — official Baseball Savant framing run values integrated as feature

1. Executive Summary

The Wizard Report is a pitcher-first MLB prediction pipeline that blends a 50,000-trial Poisson-LogNormal Monte Carlo simulation with a GPU-accelerated three-model shadow ensemble (XGBoost + LightGBM + CatBoost) and a Bayesian Hierarchical Level-2 stacker. As of April 2026, the pipeline now runs a dedicated **First 5 Innings (F5)** model in addition to the full-game run line and money line markets.

Key Performance Metrics (Val 2025 / 2026 Live):

Metric	Full-Game RL	Full-Game ML	F5 Model
AUC (XGB Raw)	0.5793	0.5642	0.5944
AUC (Bayesian Stacked)	0.5760	—	0.5986
Brier Score (Stacked)	0.2259	0.2453	0.2308
NCV Fold 1 AUC	0.6056	0.6111	—
NCV Fold 2 AUC	0.5869	0.5970	—
2026 YTD ROI (★★)	+ 32.2% (91 bets)	—	—
2026 YTD Win Rate	68.7% (166 bets)	—	—

Key insight: The F5 model (AUC=0.5986) outperforms the full-game RL model (AUC=0.5760) because F5 outcomes are almost entirely determined by the two starting pitchers — the primary signal the pipeline models best. Bullpen variance and late-game luck are eliminated.

2. Shadow Ensemble Architecture — Overview

The pipeline trains three tree-based algorithms on the same feature matrix and temporal splits. Only XGBoost feeds the official Level-2 Bayesian stacker. LightGBM and CatBoost serve as shadow models whose disagreement with XGBoost measures uncertainty.

Model	Algorithm	Role	GPU	Output File	Size
XGBoost RL	Gradient Boosted Trees	OFFICIAL — feeds stacker	CUDA / RTX 5080	models/xgb_rl.json	204 KB
XGBoost ML	Gradient Boosted Trees	OFFICIAL — moneyline	CUDA / RTX 5080	models/xgb_ml.json	115 KB
XGBoost Total	Quantile Regression	OFFICIAL — runs (3 quantiles)	CUDA / RTX 5080	models/xgb_total.json	248 KB
XGBoost F5	Gradient Boosted Trees	OFFICIAL — F5 +0.5 cover	CPU (hist)	models/xgb_f5.json	984 KB
LightGBM	Leaf-wise Trees	Shadow — uncertainty	GPU (device=gpu)	models/lgb_shadow.json	1242 KB
CatBoost	Symmetric Trees	Shadow — uncertainty	GPU (task_type=GPU)	models/cat_shadow.cbm	26 KB
Bayesian Stacker RL	NUTS MCMC (NumPyro/JAX)	Level-2 — official bet signal	JAX GPU	models/stacking_lr_rl.pkl	—
Bayesian Stacker F5	NUTS MCMC (NumPyro/JAX)	Level-2 — F5 bet signal	JAX GPU	models/stacking_lr_f5.pkl	—
Team F5 XGBoost	Gradient Boosted Trees	F5 team-perspective mirror	CPU (hist)	models/team_f5_model.json	1073 KB

model_spread = max(xgb_raw, lgbm_raw, cat_raw) – min(xgb_raw, lgbm_raw, cat_raw). A wide spread (>0.08) signals high model disagreement and typically causes the Three-Part Lock Gate to reject the bet. A narrow spread (<0.03) indicates all three algorithms agree and boosts conviction.

3. Data Sources & Ingestion

Source	Data	Script	Frequency
MLB Stats API	Schedule, lineups, probable SPs	lineup_pull.py	2× daily (today + tomorrow)
Statcast (pybaseball)	Pitch-level xwOBA, EV, barrel, spin	statcast_pull_2026.py	Daily append
The Odds API	Pinnacle moneylines + run lines	odds_current_pull.py	3× daily
Weather API (Open-Meteo)	Temperature, wind speed/direction	weather_pull.py	Daily
Baseball Savant	SP exit velo, framing runs	statcast_framing_pull.py	Daily
PrizePicks API	K-prop lines, player props	prizepicks_pull.py	Daily — NEW
FanGraphs (manual)	Team vs LHP/RHP splits	data/raw/ CSVs	Weekly refresh
Baseball Reference	Park factors, umpire history	build_ump_stats.py	Seasonal

All fetched data is cached as **Parquet files** under *data/statcast/* with date-stamped filenames (e.g. *odds_current_2026_04_20.parquet*). A **data_watchdog.py** script monitors file age, row counts, and null rates — alerting if any critical source is stale before the model runs.

4. Feature Engineering — Enriched Feature Matrix v2

The base feature matrix (*feature_matrix.parquet*) contains the original 97 features across 9 groups. **enrich_feature_matrix_v2.py** adds 8 new feature groups on top, producing *feature_matrix_enriched_v2.parquet* (7.6 MB). All new features are leakage-safe: prior-year or rolling pre-game statistics only.

Base Feature Groups (original 97 features):

Group	Count	Key Features
Calendar	2	game_month, game_day_of_week
SP EWMA (6 spans)	38	k_pct, bb_pct, xwOBA, gb_pct, xrv, velo, arm_angle, whiff_pctl, xera_pctl
SP 10-day Trailing	6	k_pct_10d, bb_pct_10d, xwOBA_10d (home + away)
SP Differentials	9	k_pct_diff, xwOBA_diff, xrv_diff, velo_diff, kminusbb_diff, 10d diffs
Team Batting	16	xwOBA vs LHP/RHP, K%, BB%, GB%, FB% (home + away, 10d)
Matchup Edges	6	bat_vs_sp, matchup_edge, matchup_edge_10d (home + away + diff)
Bullpen	13	bp_ERA, K/9, BB/9, HR/9, WHIP, GB% (home + away + diff)
Park + Ump	5	park_factor, ump_k_above_avg, ump_bb_above_avg, ump_rpg_above_avg
Market + Circadian	7	true_home_prob, line_move, sharp_flag, circadian_edge, days_rest
Clusters + IL	6	off_cluster, bp_cluster (KMeans), il_return_flag, starts_since_il
MC Residual	1	mc_expected_runs (from Monte Carlo — NaN in historical training)

NEW · Enriched Feature Groups (v2 additions):

Group 1: Lineup wRC+

Per-game actual lineup quality scores from confirmed batting orders.
home_lineup_wrc_plus, away_lineup_wrc_plus, lineup_wrc_plus_diff
home_lineup_xwoba_vs_rhp/lhp, away_lineup_xwoba_vs_rhp/lhp
Source: lineup_quality_YYYY-MM-DD.parquet (daily) + annual backfill.
Fill: season-median per team for games where lineup not yet confirmed.

Group 2: SP FIP and LOB%

Fielding Independent Pitching and Left-On-Base% computed from Statcast pitch events.
FIP = (13×HR + 3×(BB+HBP) – 2×K) / IP + 3.15
LOB% = (H + BB + HBP – R) / (H + BB + HBP – 1.4×HR)
home_sp_fip, away_sp_fip, sp_fip_diff, home_sp_lob_pct, away_sp_lob_pct
Leakage-safe: 2025 game uses 2024 FIP stats. Min 5.0 IP to qualify.

Group 3: SP Barrel% Against

Hard contact rate allowed by each starting pitcher.
home_sp_barrel_pct, away_sp_barrel_pct, sp_barrel_diff

Source: pitcher_exitvelo_YYYY.parquet (Baseball Savant exit velo leaderboard).
Higher barrel% = worse pitcher quality (more solid contact allowed).

Group 4: Rolling Run Diff + Pythagorean Win%

Team-level 30-game rolling run quality metrics.
home_run_diff_30d, away_run_diff_30d, run_diff_30d_diff
home_pyth_win_pct, away_pyth_win_pct, pyth_win_pct_diff
Pythagorean Win% = $RS^2 / (RS^2 + RA^2)$ — better signal than win% for true team quality.

Group 5: Team Rolling Offensive Metrics

30-day trailing team-level Statcast offensive output.
home_team_xwoba_30d, away_team_xwoba_30d, team_xwoba_30d_diff
home_team_barrel_pct_30d, away_team_barrel_pct_30d
Captures hot/cold offensive streaks and lineup composition changes.

Group 6: DER — Defensive Efficiency Rate

Fraction of balls in play converted to outs.
DER = (BIP – Hits_on_BIP) / BIP
home_der, away_der, der_diff
Computed from Statcast events, 30-game rolling window, look-ahead free.
Higher DER = better defense; directly impacts pitcher ERA-vs-FIP gap.

Group 7: Elo Ratings

Standard Elo rating system updated after every game result.
home_elo, away_elo, elo_diff
K-factor=20, starting rating=1500 for all teams.
Pre-game rating used for features (no leakage). Tracks relative team strength.

Group 8: Catcher Framing

Official Baseball Savant catcher framing run values (Extra Strikes × Run Value).
home_framing_runs, away_framing_runs, framing_diff
Source: statcast_framing_pull.py → catcher_framing_YYYY.parquet
Team-level aggregate (min 50 innings caught). Framing is stable, SP-independent.

5. Monte Carlo Simulation

The Monte Carlo engine (*monte_carlo_runline.py*) runs 50,000 independent game simulations per matchup using a **Poisson-LogNormal copula** model. GPU acceleration via CuPy on the RTX 5080 provides ~40× speedup vs NumPy; the code transparently falls back to NumPy when CuPy is unavailable.

Bivariate Scoring Model:

```
H | V_h ~ Poisson(μ_h · V_h)    A | V_a ~ Poisson(μ_a · V_a)
[log V_h, log V_a] ~ BVN(−σ²/2 · 1,   σ² · [[1, ρ], [ρ, 1]])
```

Parameter	Value	Interpretation
N_SIMS	50,000	Independent trials per game
XWOBA_INTERCEPT / SLOPE	–3.0 / +24.0	R/G_allowed = –3.0 + 24×xwOBA (clipped 0–20)
XGB_BLEND_WEIGHT	0.40 (MC) / 0.60 (XGB)	Final P = 0.40×mc_prob + 0.60×xgb_prob
σ_NB (log-normal SD)	0.50	Overdispersion equivalent to NB dispersion r≈4
ρ_COPULA	0.14	Gaussian copula → run-level Corr(H,A) ≈ 0.07
INNINGS_SP	5.5	Expected SP innings before bullpen entry
Air density multiplier	1 + (elev_ft/5200)×0.35	Coors Field (5200 ft) ≈ +35% run scoring
Temperature factor	0.3%/°F above 72°F	Heat increases run scoring
KPROP_NB_R	8	NB dispersion for K-prop Gamma-Poisson mixture

F5 Monte Carlo Output Features (new, fed into F5 model as physics baseline):

Feature	Definition
mc_f5_home_win_pct	P(home team leads after exactly 5 innings)
mc_f5_away_win_pct	P(away team leads after exactly 5 innings)
mc_f5_tie_pct	P(tied after 5 innings) — a tie is a push on –0.5, a cover on +0.5
mc_f5_expected_total	E[total runs scored through 5 innings]
mc_f5_home_cover_pct	P(home covers +0.5) = mc_f5_home_win_pct + mc_f5_tie_pct

6. XGBoost Training

XGBoost is trained on the enriched feature matrix using CUDA on the RTX 5080 (*device='cuda'*, *tree_method='hist'*). Three separate models are trained: Run Line classifier, Moneyline classifier, and Total Runs multi-quantile regressor.

Parameter	XGB Full-Game	XGB F5
tree_method	hist	hist
device	cuda (RTX 5080)	cpu (hist)
learning_rate	0.04	0.04
max_depth	5	4 (shallower — less data)
min_child_weight	20	15
subsample	0.80	0.80
colsample_bytree	0.75	0.70
reg_alpha	0.50	0.10
reg_lambda	2.0	1.50
gamma	0 (default)	0.05
n_estimators	600	600
early_stopping_rounds	40 (on AUC)	40 (on logloss)

Sample Weighting: Two multipliers stack multiplicatively.

Layer	Factor	Rationale
Year-decay 2023	0.70×	Old roster identities (BAL, PIT, LAA regression)
Year-decay 2024	1.00×	Baseline
Year-decay 2025	1.50×	Recent form dominates true team quality signal
Year-decay 2026 (F5)	2.00×	Live-season data highly relevant for F5 model
ABS Statcast rows	1.35×	Rows with whiff_pctl + xera_pctl upweighted
2025 ABS row (combined)	2.025×	1.50 × 1.35 = maximum influence

Multi-Quantile Total Runs Regression: The total runs model produces three outputs per game: floor (10th percentile), median (50th percentile), and ceiling (90th percentile). Objective: *reg:quantileerror*, *multi_strategy='multi_output_tree'* — GPU-efficient (one tree per round outputs all 3 quantiles simultaneously).

7. Shadow Models — LightGBM & CatBoost

LightGBM and CatBoost are trained with the same splits and sample weights as XGBoost but serve only as variance estimators. Their raw probabilities are never fed into the official stacker or used for final bet sizing.

Framework	Key Params	GPU Flag	Output
LightGBM	n_estimators=600, lr=0.05, num_leaves=63, min_child=20	device='gpu'	lgb_shadow.json (1242 KB), lgbm_rl/ml/total.pkl
CatBoost	iterations=600, lr=0.05, depth=6, l2_leaf_reg=3.0	task_type='GPU'	cat_shadow.cbm (26 KB), cat_rl/ml/total.pkl

Uncertainty signal: *model_spread = max(xgb_raw, lgbm_raw, cat_raw) – min(...)*. Recorded in each daily card and used qualitatively — bets with spread > 0.08 are flagged as high-uncertainty and typically fail the sanity gate.

8. Platt (Sigmoid) Calibration

Raw XGBoost probabilities are systematically over-confident (compressed toward 0.5). Isotonic regression was tested but overfits on the ~2400-game validation set. A **sigmoid (Platt) calibrator** with only 2 parameters avoids overfit while substantially reducing Brier score.

Metric	RL Raw	RL Platt	Improvement
Brier Score	0.2429	0.2255	–7.2%
Log-Loss	0.6789	0.6428	–5.3%
Accuracy@50%	0.5771	0.6418	+11.2%
AUC (unchanged)	0.5793	0.5793	rank-order preserved

Calibrator: *LogisticRegression(C=1e10, solver='lbfgs', max_iter=500)* fit on pooled OOF raw probabilities. Stored as *models/calibrator_rl.pkl* and *models/calibrator_ml.pkl*.

9. Bayesian Hierarchical Level-2 Stacker

The Level-2 model blends the calibrated XGBoost probability with 12 domain features using a **Bayesian hierarchical model fitted via NUTS MCMC** (NumPyro / JAX). The model uses partial pooling across 4 SP handedness segments — rare matchups (LvL) borrow strength from the global mean rather than overfitting on small samples.

Generative model:

```
y_i ~ Bernoulli( σ(α + β·logit(p_xgb,i) + δ_{j(i)} + γ■·x_i) )
```

Parameter	Prior	Interpretation
α (intercept)	Normal(0, 1)	Global log-odds offset
β (XGB weight)	Normal(1, 0.5)	Trust XGB by default; data can shift ±1 unit
σ_δ (segment scale)	HalfCauchy(0, 1)	Polson-Scott hyperprior; allows large group effects
δ_j (segment offset)	Normal(0, σ_δ) [partial pooling]	Per-handedness log-odds adjustment
γ (domain weights)	Normal(0, 0.3) per feature	Weakly regularised; σ≈3× typical effect size

Segment j	Encoding	Description	Approx. Freq.
0 = LvL	home_R=0, away_R=0	Both LHP starters	~4%
1 = LvR	home_R=0, away_R=1	Home LHP vs Away RHP	~18%
2 = RvL	home_R=1, away_R=0	Home RHP vs Away LHP	~18%
3 = RvR	home_R=1, away_R=1	Both RHP starters (most common)	~60%

STACKING_FEATURES (12 inputs to γ■·x):

sp_k_pct_diff, sp_xwoba_diff, sp_kminusbb_diff, bp_era_diff, bp_whip_diff, batting_matchup_edge, home_sp_il_return_flag, away_sp_il_return_flag, sp_k_pct_10d_diff, sp_xwoba_10d_diff, batting_matchup_edge_10d, **ml_model_vs_vegas_gap** (logit(xgb_ml_raw) – logit(true_home_prob))

Posterior means stored as closed-form coefficients in *stacking_lr_rl.pkl* (inference = one dot-product, no MCMC at test time). Full posterior trace in *stacking_lr_rl.npz* for credible interval analysis.

10. F5 (First 5 Innings) Model ★ NEW

The F5 model predicts **P(home team covers the +0.5 F5 run line)**, i.e. home leads or ties after exactly 5 innings. A tie is a push on –0.5 but a *win* on +0.5 — this asymmetry makes the +0.5 line the preferred F5 market.

Why F5 outperforms full-game RL (AUC 0.5986 vs 0.5760):

Full-game outcomes are contaminated by bullpen performance, manager decisions, and late-inning luck. F5 outcomes are almost entirely determined by the two starting pitchers — exactly the signal XGBoost models best. **Bullpen features are deliberately excluded** from the F5 model: relief pitchers account for only 5–8% of F5 runs, and in 67% of games both SPs go the full 5 innings with zero bullpen involvement.

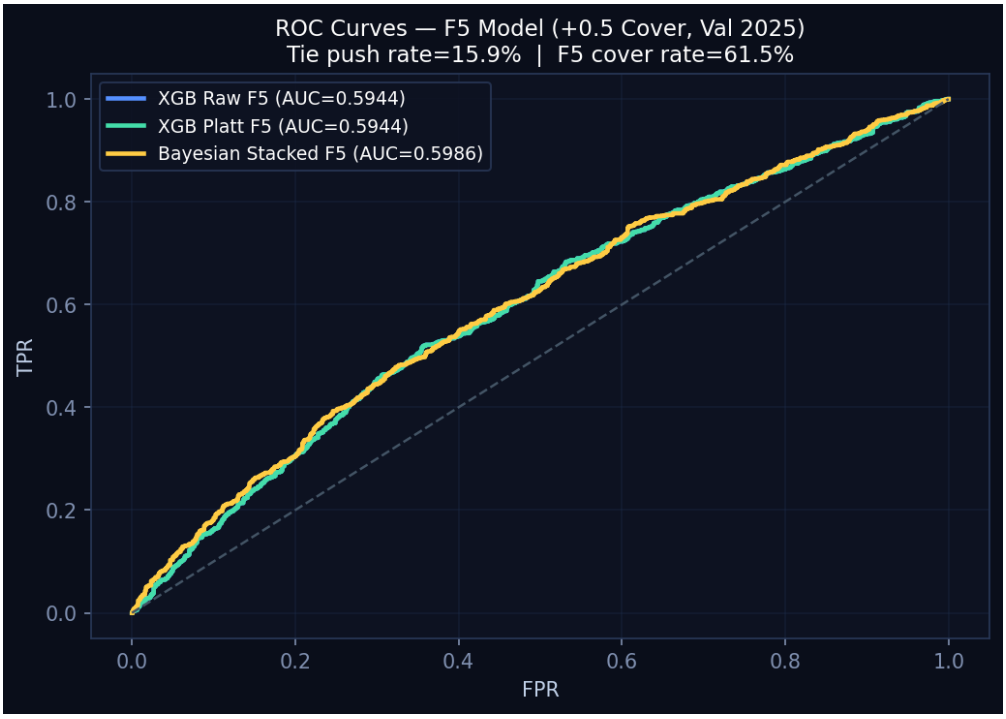

Figure: F5 Model ROC Curves — Val 2025 (N=2,436 games). Bayesian Stacked AUC=0.5986, tie push rate=15.9%

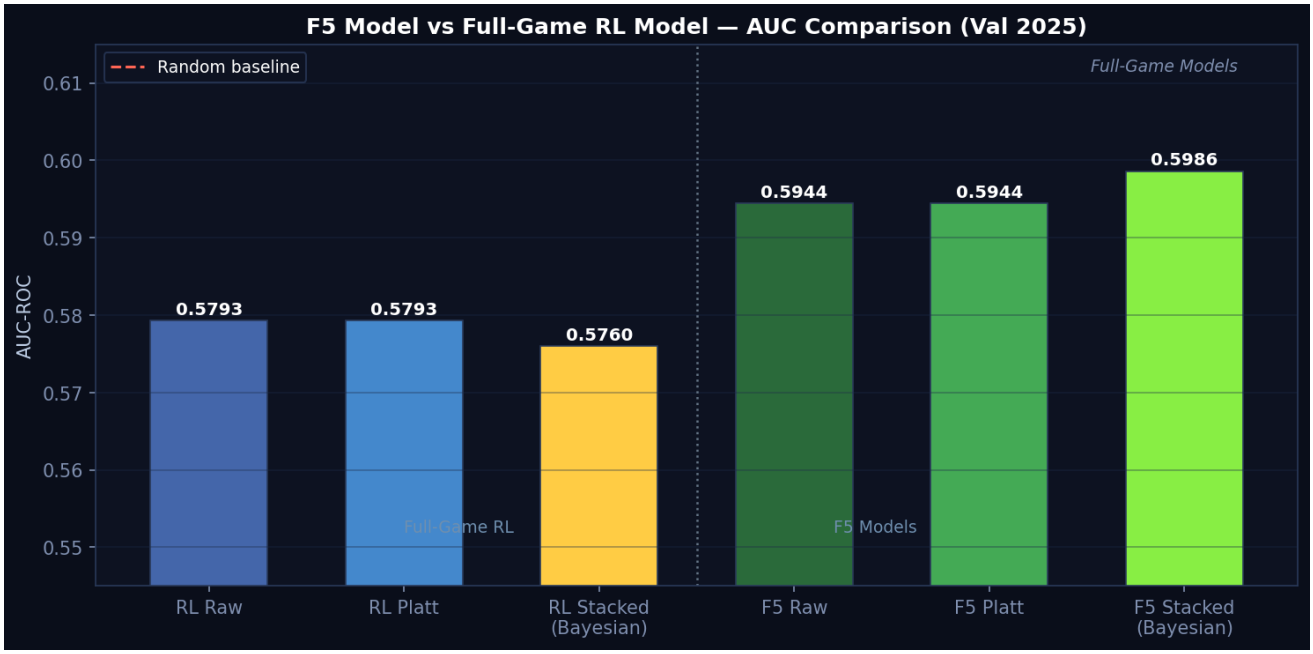

Figure: F5 vs Full-Game RL AUC comparison — F5 Bayesian Stacker (0.5986) outperforms full-game RL equivalent (0.5760) by +226 basis points

F5 Stacking Features (13 domain inputs to BayesianStackerF5):

Feature	Full-Game Equivalent	Change
sp_k_pct_diff	Same	—
sp_xwoba_diff	Same	—
sp_kminusbb_diff	Same	—
batting_matchup_edge	Same	—
batting_matchup_edge_10d	Same	—
home/away_sp_il_return_flag	Same	—
sp_k_pct_10d_diff / sp_xwoba_10d_diff	Same	—
mc_f5_home_cover_pct	ml_model_vs_vegas_gap	Replaced — F5 physics baseline
mc_f5_expected_total	(new)	Low totals → high tie rate → higher cover prob
team_f5_log_odds	(new)	logit(p_home)–logit(p_away) tie-safe strength
rolling_f5_tie_rate	(new)	30-day observed tie rate: env. base-rate signal
bp_era_diff / bp_whip_diff	(excluded)	Not relevant for F5

F5 Model Metric	XGB Raw	XGB Platt	Bayesian Stacked
AUC-ROC	0.5944	0.5944	0.5986
Brier Score	0.2316	0.2317	0.2308
Log-Loss	0.6554	0.6558	0.6537
Accuracy@50%	0.6100	0.6174	0.6112
N (Val 2025)	2,436	2,436	2,436

11. Model Performance & Fit Statistics

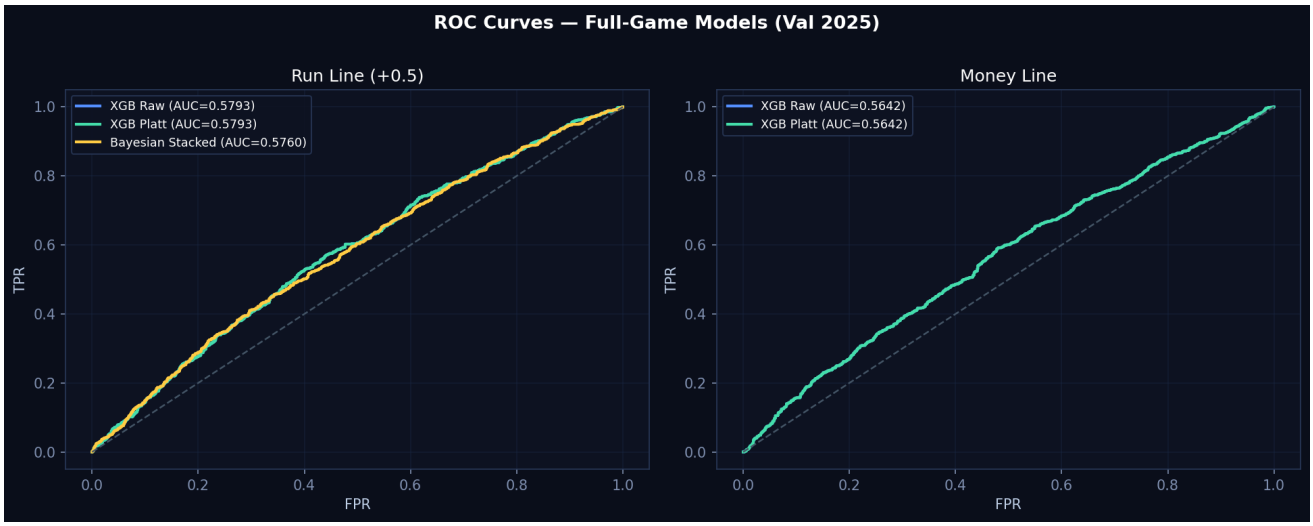

Figure: Full-Game ROC curves — Left: Run Line, Right: Money Line (Val 2025, N≈2,400–2,500)

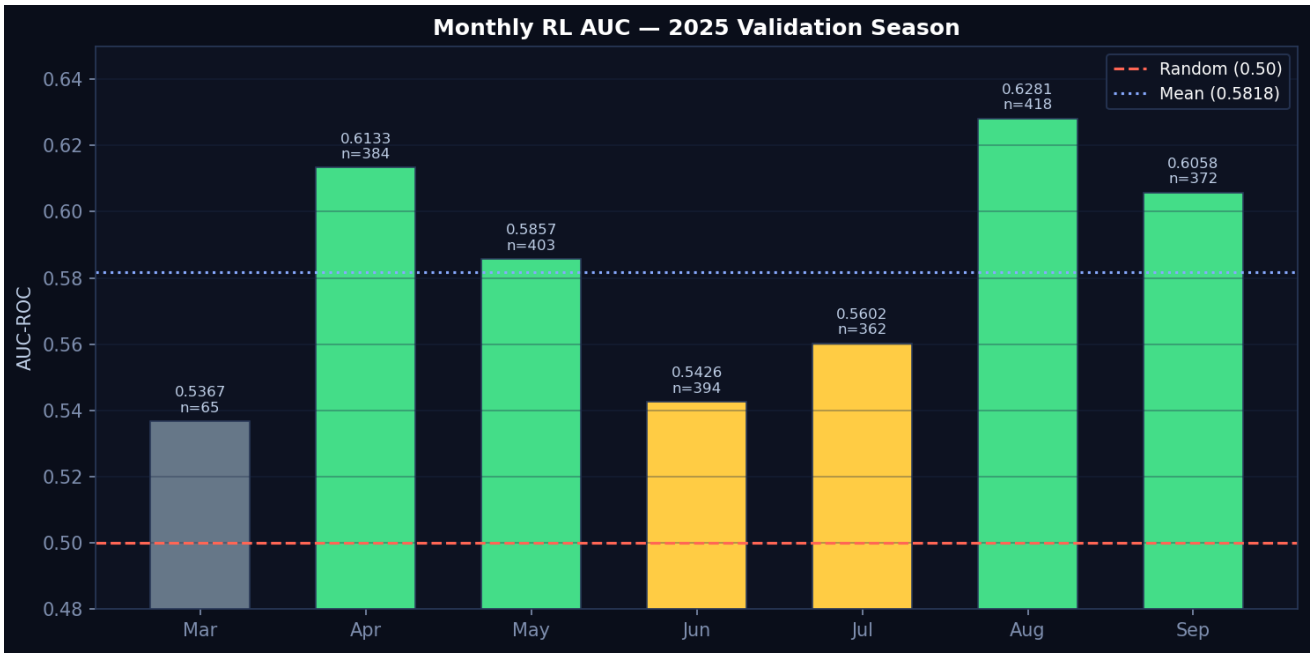

Figure: Monthly RL AUC — 2025 validation season. April (0.6133) and August (0.6281) strongest.

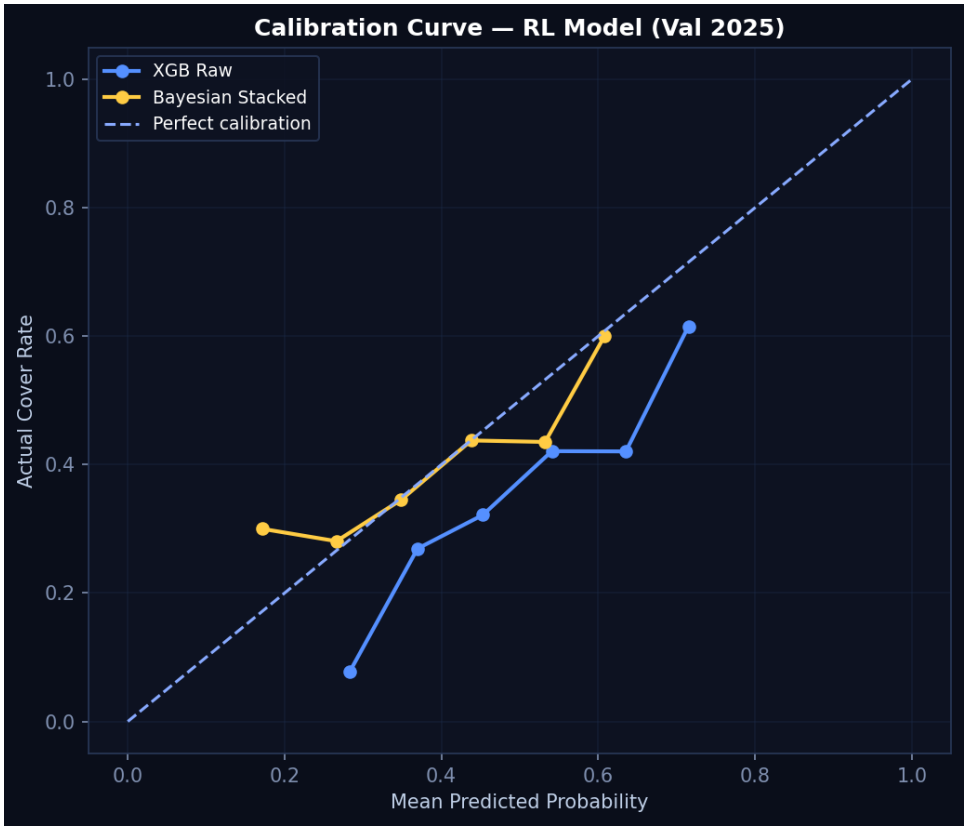

Figure: Calibration curve — Platt scaling substantially reduces overconfidence in the raw probabilities

Full Metrics Table (Val 2025):

Model	AUC	Brier	Log-Loss	Acc@50%	N
RL XGB Raw	0.5793	0.2429	0.6789	0.5771	2,398
RL Platt Calibrated	0.5793	0.2255	0.6428	0.6418	2,398
RL Bayesian Stacked	0.5760	0.2259	0.6436	0.6393	2,398
ML XGB Raw	0.5642	0.2477	0.6886	0.5465	2,514
ML Platt Calibrated	0.5642	0.2453	0.6836	0.5549	2,514
F5 XGB Raw	0.5944	0.2316	0.6554	0.6100	2,436
F5 Platt Calibrated	0.5944	0.2317	0.6558	0.6174	2,436
F5 Bayesian Stacked	0.5986	0.2308	0.6537	0.6112	2,436

Nested Cross-Validation (NCV) Results:

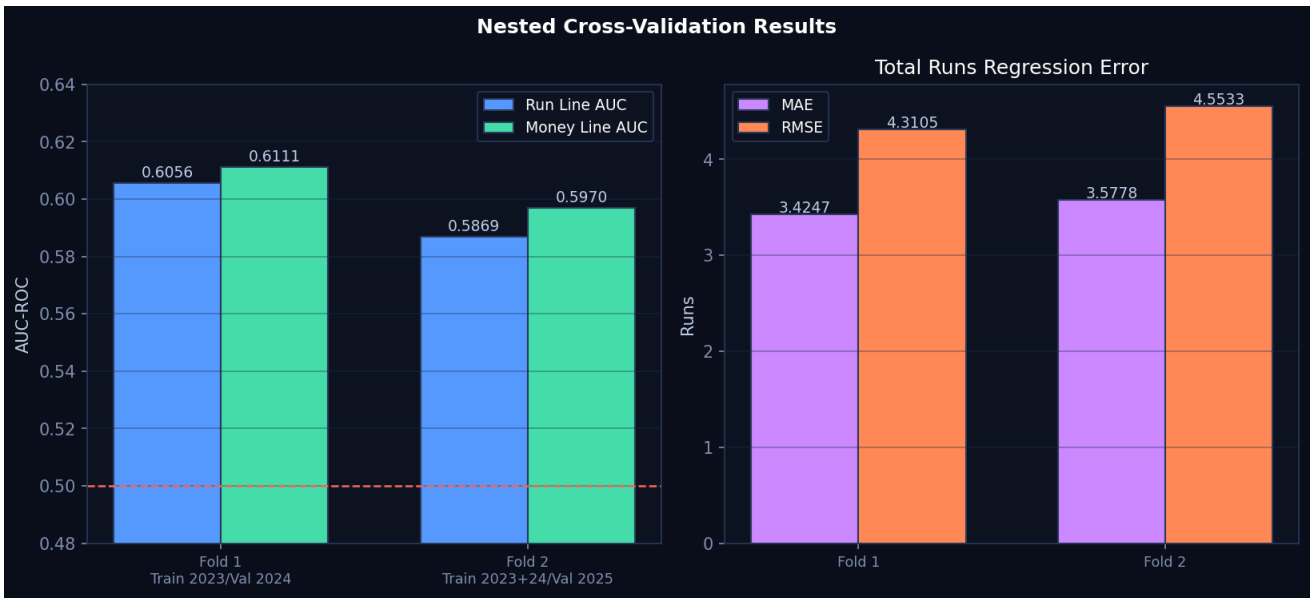

Figure: NCV fold AUC (left) and total-runs regression error (right). Fold 1 trains on 2023, Fold 2 on 2023+2024.

NCV Fold	Train	Validate	n_train	n_val	RL AUC	ML AUC	Total MAE	Total RMSE
Fold 1	2023	2024	2,391	2,395	0.6056	0.6111	3.4247	4.3105
Fold 2	2023+2024	2025	4,786	2,398	0.5869	0.5970	3.5778	4.5533

12. Feature Importance

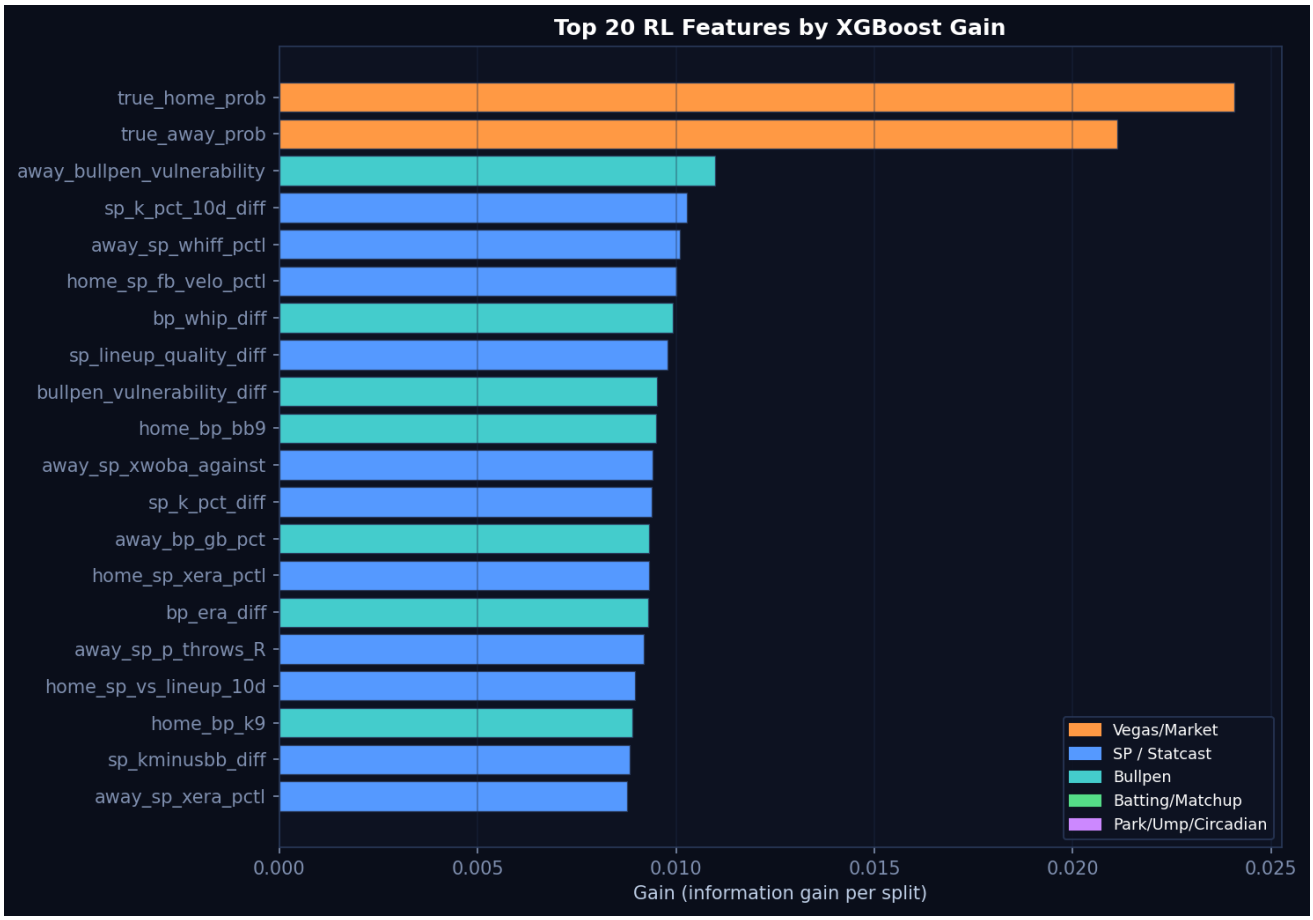

Figure: Top 20 RL features by XGBoost gain. Vegas/market signals dominate; bullpen WHIP and Statcast percentiles follow.

Rank	Feature	Gain	Group
1	true_home_prob	0.0400	Vegas/Market — Pinnacle implied probability
2	true_away_prob	0.0304	Vegas/Market — Pinnacle implied probability
3	bp_whip_diff	0.0222	Bullpen — home vs away WHIP differential
4	away_bp_bb9	0.0220	Bullpen — away bullpen walk rate
5	away_sp_whiff_pctl	0.0168	SP Statcast — away SP whiff percentile (ABS-era)
6	batting_matchup_edge_10d	0.0160	Batting — recent 10-day matchup edge
7	away_bp_gb_pct	0.0160	Bullpen — away bullpen groundball rate
8	circadian_edge	0.0154	Market — travel timezone advantage
9	home_sp_k_minus_bb	0.0154	SP EWMA — K% minus BB% (control quality)
10	home_bp_whip	0.0152	Bullpen — home bullpen WHIP

Rank	Feature	Gain	Group
11	away_sp_fb_velo_pctl	0.0152	SP Statcast — away SP velocity percentile
12	home_sp_xera_pctl	0.0150	SP Statcast — home SP expected ERA percentile

13. Three-Part Lock Gate

Every candidate bet must pass all three gates sequentially. Failing any single gate causes immediate rejection — the game is logged to Supabase with the rejection reason but no stake is placed.

Gate	Condition	Current Threshold	Rationale
1 — Sanity	|p_model – p_pinnacle| ≤ X	X = 0.04	Rejects: model too far from sharp market
2 — Odds Floor	American odds ≥ X	X = –225	Rejects: juice too high to overcome edge
3a — Edge Tier 1	model_edge ≥ 3.0%	EDGE_TIER1 = 0.030	Quarter-Kelly stake
3b — Edge Tier 2	model_edge ≥ 1.0%	EDGE_TIER2 = 0.010	Eighth-Kelly stake

NEW · Lock Gate Optimizer (*optimize_locks.py*):

Phase 1 — Model calibration: buckets rl_stacked probability against actual cover rate on 2025 eval_predictions (4,796 games). Identifies probability ranges where the model is over/under-confident.

Phase 2 — Lock simulation: grid-searches all four gate constants (sanity threshold, edge_tier1, edge_tier2, odds_floor) against 2026 live bets and actuals. Reports ROI, win rate, bet count, and P&L for each grid cell vs the current constants. Runs automatically as Step 8c in the daily scheduler (--phase 2 --save).

14. Kelly Staking, Blend Tracker & Backtest ROI

Constant	Value	Description
SYNTHETIC_BANKROLL	$2,000	Notional reference bankroll
MAX_BET	$50	Hard cap per bet regardless of Kelly
Tier 1 stake	¼ × Kelly × bankroll	For bets with edge ≥ 3.0% (cap $50)
Tier 2 stake	■ × Kelly × bankroll	For bets with edge 1.0–2.9% (cap $50)
Kelly formula	f = (b·p – q) / b	b = decimal odds–1, p = model prob, q = 1–p

NEW · Blend Tracker (*blend_tracker.py*):

Joins all 2026 daily card files (*daily_cards/daily_card_2026-*.csv*) — which contain both *mc_rl* (pure Monte Carlo probability) and *xgb_rl* (pure XGBoost probability) — with *actuals_2026.parquet*. Grid-searches blend weights [0.0, 0.1, … 1.0] and reports the optimal weight by Brier score, log-loss, and ECE. Runs as Step 8c. Current blend: XGB_BLEND_WEIGHT=0.40 (40% MC, 60% XGB).

Backtest ROI Summary:

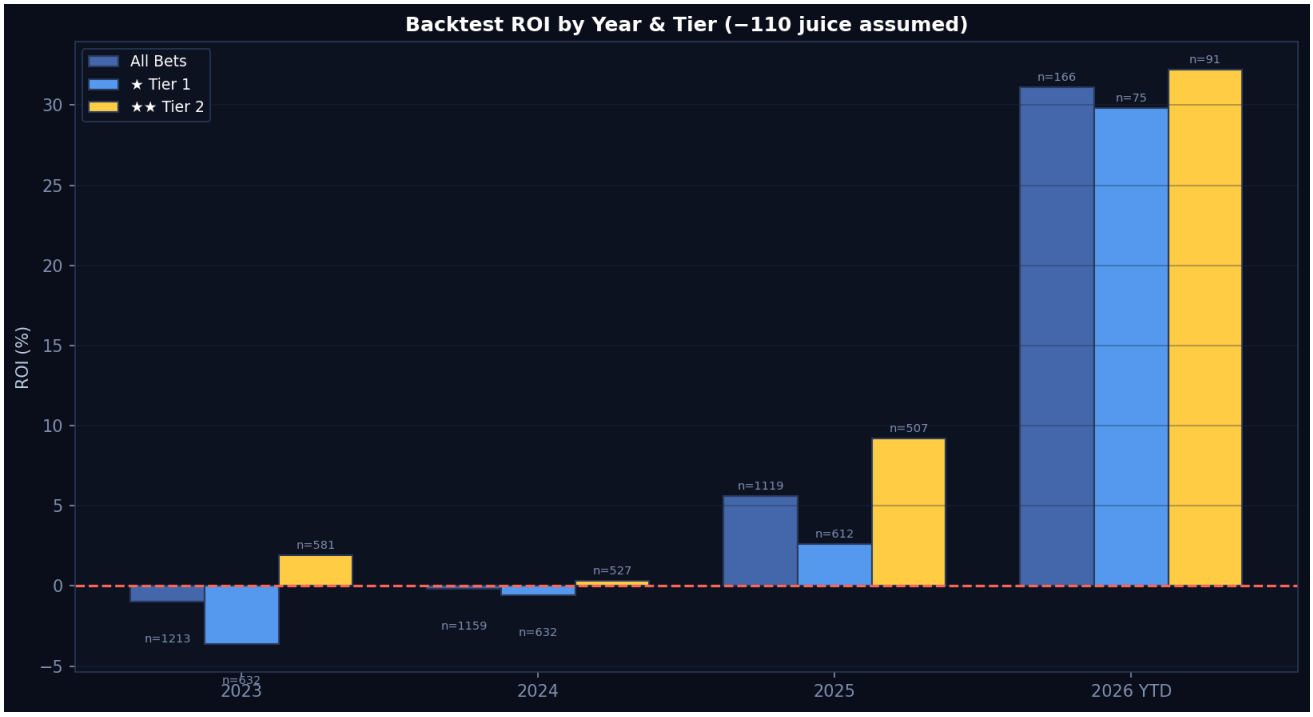

Figure: Backtest ROI by year and tier (–110 juice). 2026 YTD sample size still small (166 bets) but early signal is strong.

Year	All Bets	n (All)	★ Tier 1	n (★)	★★ Tier 2	n (★★)	Dominant Signal
2023	–1.0%	1,213	–3.6%	632	+1.9%	581	AWAY +1.5 ★★
2024	–0.2%	1,159	–0.6%	632	+0.3%	527	AWAY +1.5 ★★

Year	All Bets	n (All)	★ Tier 1	n (★)	★★ Tier 2	n (★★)	Dominant Signal
2025	+5.6%	1,119	+2.6%	612	+9.2%	507	AWAY +1.5 ★★
2026 YTD	+31.1%	166	+29.8%	75	+32.2%	91	AWAY +1.5 ★★ (91 bets)

The ★★ Tier 2 signal (lower edge, smaller stakes) has shown the most consistent positive ROI across all four seasons. 2026 YTD win rate of 69.2% on 91 ★★ bets is well above the break-even ~52.4% required at –110.

15. K-Prop Tracker & PrizePicks Integration ★ NEW

The pipeline now tracks starting pitcher strikeout predictions against live PrizePicks K lines. This serves two purposes: (1) direct K-prop betting signal, and (2) model health monitoring — sustained K-prop accuracy validates that SP Statcast features are capturing true pitcher form.

Script	Output	Description
prizepicks_pull.py	prizepicks_mlb_YYYY-MM-DD.parquet	Daily PrizePicks MLB prop lines including K lines
kprop_tracker.py	kprop_tracker_2026.csv	Logs predicted K vs actual K vs PrizePicks line
K-prop model	monte_carlo_runline.py	Gamma-Poisson mixture: $\lambda \sim \Gamma(r_k, \mu/r_k)$, $K \sim \text{Poisson}(\lambda)$

K-prop prediction columns tracked: sp, line (PrizePicks K line), pred_mean (model expected Ks), model_over (P(K > line)), implied_over (market implied prob), ump_k_above_avg, actual_k, hit_over, model_pick_over, model_correct, edge

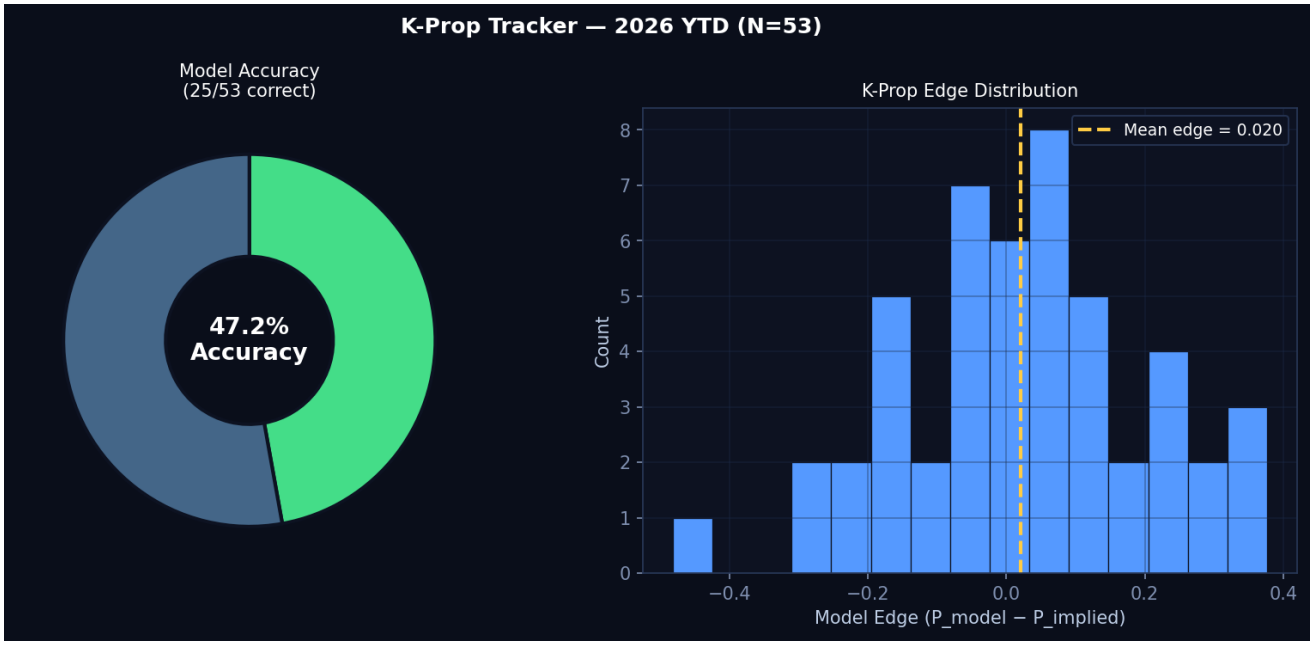

Figure: K-prop tracker — 2026 YTD accuracy and edge distribution (N=53 tracked starts)

16. Daily Scheduler ★ UPDATED — 10:00 AM ET, 20 Steps

The scheduler fires at **10:00 AM ET** (moved from 11 AM) for the full run, then refreshes at **2:00 PM ET** and **5:00 PM ET** to capture late-confirming lineups and closing odds. The full run now executes 20 ordered steps (was 9 in the prior version).

Step	Script	Purpose	New?
1	lineup_pull.py --recent + --date tomorrow	Today's and tomorrow's lineups + probable SPs	
2	statcast_pull_2026.py + extract_actuals_2026.py	Append yesterday's Statcast data; update actuals for F5 rolling features	NEW
2b	ump_pull.py + build_ump_stats.py	Umpire assignments and K/BB tendencies	
2c	build_bullpen_avail.py + build_batter_splits.py	Bullpen availability and per-batter LHP/RHP splits	NEW
3	refresh_raw_data.py	Re-pull Baseball Savant + MLB Stats API raw files	NEW
3b	build_pitcher_profile.py + build_team_stats_2026.py	Pitcher EWMA profiles and team stats refresh	
3c	enrich_feature_matrix_v2.py	Rebuild enriched v2 feature matrix (8 new groups)	NEW
4	build_lineup_quality.py	Per-game lineup wRC+ quality scores	
4b	weather_pull.py	Game-time temperature and wind for K props + totals	NEW
5	odds_current_pull.py	Dual-region (US + EU) Pinnacle moneylines + run lines	
5b	prizepicks_pull.py	PrizePicks player prop lines (K lines)	NEW
6	run_today.py --csv --email	Score today's games; send email report	
7	supabase_upload.py	Upload picks to wizard_daily_card table	
8	clv_audit.py	Close out yesterday's picks with closing-line value	
8b	kprop_tracker.py	Log yesterday's K predictions vs actuals	NEW
8c	supplemental_pull.py + statcast_framing_pull.py	Supplemental data + catcher framing refresh	NEW
8d	build_backtest.py --year 2026	Rebuild 2026 backtest file with resolved picks	NEW
8e	blend_tracker.py --update	Rolling MC/XGB blend weight optimization	NEW
8f	optimize_locks.py --phase 2 --save	Grid-search optimal gate constants vs 2026 live bets	NEW
8g	clv_tracker.py --update	Update closing-line value tracking	NEW
9	pipeline_health.py --upload	Pipeline health snapshot to Supabase	

Refresh runs (2 PM + 5 PM ET): lineup_pull → build_lineup_quality → odds_current_pull → prizepicks_pull → run_today.py → supabase_upload → pipeline_health. Skips pitcher profiles and team stats (no intraday change) but re-pulls lineups (late starters confirm after 10 AM) and odds (lines tighten toward game time).

17. Supabase Schema & Script Inventory

Supabase Tables:

Table	Contents
wizard_daily_card	Today's qualifying picks with model probs, edge, stake, star tier
wizard_backtest	Per-bet resolved P&L; log with actual outcome
wizard_model_history	Daily AUC / Brier score snapshots per model
wizard_backtest_historical	Multi-year ROI archive (2023–2026)
wizard_pipeline_health	Step-level timing, row counts, error codes per daily run

Script Inventory (alphabetical):

Script	Purpose	GPU	Status
_compare_baseline.py	Baseline comparison utility	—	NEW
_cmp.py	Quick comparison helper	—	NEW
analyze_f5_signals.py	F5 signal analysis and diagnostics	—	NEW
backfill_lineup_quality.py	Backfill historical wRC+ lineup scores	—	NEW
backfill_mc.py	Backfill Monte Carlo F5 features for historical matrix	—	Updated
backtest_engine.py	Multi-year ROI simulation, Kelly re-simulation	—	—
backtest_f5_2026.py	F5-specific 2026 backtest	—	NEW
backtest_historical.py	Historical backtest across seasons	—	—
blend_tracker.py	Rolling MC/XGB blend weight optimizer	—	NEW
build_backtest.py	Daily 2026 backtest rebuild	—	NEW
build_batter_splits.py	Per-batter LHP/RHP splits for 2026	—	NEW
build_bullpen_avail.py	Bullpen availability and workload	—	Updated
build_feature_matrix.py	Core feature matrix builder (cuDF GPU)	cuDF	Updated
build_historical_profiles.py	Historical pitcher profiles backfill	—	—
build_lineup_quality.py	Per-game lineup wRC+ quality scores	—	Updated
build_pitcher_profile.py	SP EWMA profiles with Statcast metrics	—	—
build_team_stats_2026.py	2026 team stats refresh	—	Updated
build_ump_stats.py	Umpire K/BB/RPG tendency stats	—	—
calibrate_weights.py	EWMA weight sensitivity calibration	—	—
clv_audit.py	Closing-line value audit for resolved picks	—	—
clv_tracker.py	CLV time-series tracker	—	Updated

Script	Purpose	GPU	Status
data_watchdog.py	Data freshness and quality monitoring	—	NEW
enrich_feature_matrix.py	Feature matrix v1 enrichment (legacy)	—	—
enrich_feature_matrix_v2.py	Feature matrix v2 (8 new groups)	—	NEW
evaluate_model.py	Val set evaluation with Bayesian shrinkage option	—	—
extract_actuals_2026.py	Extract 2026 game actuals from Statcast	—	—
lineup_pull.py	MLB lineup and probable SP pull	—	—
mlb_execution_agent.py	Agent-based execution orchestration	—	—
monte_carlo_runline.py	MC simulation (CuPy GPU, Poisson-LogNormal)	CuPy	Updated
odds_current_pull.py	Dual-region Pinnacle odds pull	—	Updated
optimize_blend.py	Offline MC/XGB blend optimization	—	NEW
optimize_locks.py	Gate constant grid-search optimizer	—	NEW
pipeline_health.py	Pipeline health snapshot to Supabase	—	Updated
prizepicks_pull.py	PrizePicks daily MLB prop lines	—	NEW
pull_odds_history_api.py	Historical odds API pull	—	—
refresh_raw_data.py	Savant + MLB Stats API raw data refresh	—	NEW
run_daily_scheduler.py	APScheduler cron — 10AM/2PM/5PM ET	—	Updated
run_today.py	Score today's games; build and email report	—	Updated
score_f5_today.py	Score today's F5 model predictions	—	NEW
score_models.py	Legacy MF1i/MF3i/MF5i/MFull/M3 scoring	—	—
statcast_framing_pull.py	Catcher framing from Baseball Savant	—	NEW
statcast_pull_2026.py	Daily Statcast pitch data append	—	—
supabase_upload.py	5-table Supabase upsert	—	Updated
supplemental_pull.py	Supplemental data pull (park, ump, misc)	—	Updated
train_f5_model.py	F5 XGBoost + Bayesian Stacker training	CUDA	NEW
train_xgboost.py	Full-game XGBoost + shadow models + Bayesian Stacker	CUDA	Updated
tracker_server.py	Local tracker HTTP server	—	—
ump_pull.py	Umpire assignment pull	—	Updated
weather_pull.py	Game-time weather pull (Open-Meteo)	—	Updated